UET NOWSHERA · Electrical Engineering · Computer Programming

LAB REPORT 2

Basic Syntax and Input/Output in C++

Computer Programming · UET NOWSHERA · Electrical Engineering

1 Objective

The main objective of this lab is to understand the basic structure of a C++ program. In this lab, we learn how to use input and output statements, work with different data types, write programs with proper indentation, and perform calculations using arithmetic operators.

2 Theoretical Background

Every C++ program follows a specific structure. It usually includes header files, the **main()** function, and program statements. The program execution always starts from the **main()** function.

The **iostream** library is used for input and output operations. The **cout** statement displays output on the screen, while **cin** takes input from the user. The symbols **<<** and **>>** are called stream insertion and extraction operators.

C++ is a statically typed language — every variable must be declared with a specific data type before use. Common data types include:

- int — used for integer numbers
- double — used for decimal numbers
- char — used for a single character
- bool — used for true or false values
- string — used to store text

C++ provides arithmetic operators such as +, –, *, /, and % for mathematical calculations. Good indentation makes programs easier to read. Comments are written using **//** for single-line and **/* */** for multi-line.

3 Software Used

IDE	DEV-C++ IDE
COMPILER	C++ Compiler (included in DEV-C++)

4 Experimental Programs

Program 1 — Hello World and User Greeting

A C++ program was written that asks the user to enter their name and then displays a personalised greeting. This demonstrates the use of **string** variables, **cin** for input, and **cout** for output.

Algorithm

- Start the program
- Declare a variable to store the user's name
- Ask the user to enter their name
- Display a greeting message
- End the program

C++ Code

```
#include <iostream>
using namespace std;

int main()
{
    string name;
    cout << "Enter your name: ";
    cin >> name;
    cout << "Hello " << name << "! Welcome to C++ programming.";
    return 0;
}
```

Program 2 — Simple Calculator Using Arithmetic Operators

A calculator program was written that takes two integer inputs from the user and displays the results of addition, subtraction, multiplication, and division.

Algorithm

- Start the program
- Declare two integer variables
- Take input from the user
- Perform addition, subtraction, multiplication, and division
- Display the results
- End the program

C++ Code

```
#include <iostream>
using namespace std;

int main()
{
    int a, b;
    cout << "Enter two numbers: ";
    cin >> a >> b;
    cout << "Addition       = " << a + b << endl;
    cout << "Subtraction    = " << a - b << endl;
    cout << "Multiplication = " << a * b << endl;
    cout << "Division       = " << a / b << endl;
    return 0;
}
```

5 Conclusion

In this lab, we learned about the basic structure of a C++ program. We also learned how to use input and output statements, work with different data types, and perform mathematical calculations using arithmetic operators. This lab helped us understand the fundamental concepts of C++ programming, which are essential for writing more complex programs in the future.